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30 March 1983
SUPERSEDING
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14 November 1960

# MILITARY SPECIFICATION

## STAIRCASE, AIRCRAFT BOARDING, MOBILE, TYPE MA-1

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of mobile, aircraft boarding staircase, designated type MA-1.

## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2.), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | | | |
|---|---|---|---|
| | MIL-T-704 | - | Treatment and Painting of Material. |
| | MIL-F-3541 | - | Fittings, Lubrication. |
| | MIL-C-4751 | - | Casters, Rigid and Swivel, Precision, Heavy Duty, and Shock Absorbing. |
| * | MIL-W-5044 | - | Walkway, Compound, Nonslip and Walkway Matting, Nonslip. |
| | MIL-W-5050 | - | Walkway, Coating and Matting, Nonslip, Aircraft Application of. |
| * | MIL-P-5315 | - | Packing, Performed, Hydrocarbon Fuel Resistant. |
| | MIL-H-5606 | - | Hydraulic Fluid, Petroleum Base, Missile and Ordnance. |
| * | MIL-M-8090 | - | Mobility, Towed Aerospace Ground Equipment, General Requirements for. |
| | MIL-A-8421 | - | Air Transportability Requirements, General Specification for. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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| | | | |
|---|---|---|---|
| | DOD-D-1000 | - | Drawing, Engineering and Associated Lists. |
| * | MIL-L-10547 | - | Liner, Case, and Sheet Overwrap, Water Vapor-proof or Waterproof Flexible. |
| | MIL-B-11188 | - | Batteries, Storage, Lead-Acid (Starting, Lighting, and Ignition-Automotive Type). |
| * | MIL-H-81019 | - | Hydraulic Fluid, Petroleum Base, Ultra-low Temperature. |

STANDARDS

FEDERAL

| | | | |
|---|---|---|---|
| * | FED-STD-H28 | - | Screw Thread Standard for Federal Services. |
| * | FED-STD-595 | - | Color |

MILITARY

| | | | |
|---|---|---|---|
| | MIL-STD-129 | - | Marking for Shipment and Storage. |
| | MIL-STD-130 | - | Identification Marking of U.S. Military Property. |
| * | MIL-STD-162 | - | Materials Handling Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Process. |
| * | MIL-STD-794 | - | Parts and Equipment, Procedures for Packaging of. |
| * | MIL-STD-810 | - | Environmental Test Methods. |
| * | MIL-STD-831 | - | Test Report, Preparation of. |
| * | MIL-STD-1188 | - | Commercial Packaging of Supplies and Equipment. |
| | MS24373 | - | Caster, Rigid and Swivel, Precision Pneumatic Tired. |

AIR FORCE - NAVY AERONAUTICAL

| | | | |
|---|---|---|---|
| | AN6227 | - | Packing - "O" Ring Hydraulic. |

DRAWINGS

ORDNANCE CORPS

| | | | |
|---|---|---|---|
| | B7041347 | - | Lunette Eye. |

AIR FORCE

| | | | |
|---|---|---|---|
| | 49B6412 | - | Ring - Packing, Back Up. |
| | 49B6413 | - | Ring - Packing, Back Up. |

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, Tarif Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing (see 4.3).

3.2 Components. The staircase shall consist of the following major components:

a. Lift mechanism.

b. Lift lock.

c. Running gear.

d. Immobilization jacks.

e. Handrails.

f. Electrical system.

g. Steps and platform.

h. Tow bar.

3.3 Materials.

3.3.1 Selection of materials. Specifications and standards for all materials, parts and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with procedures established by the procuring activity or as specified in the contract or order, except as provided in the following paragraph.

3.3.1.1 Standard parts. Standard parts (MS or AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

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3.3.2 Protective treatment. When materials are used in the construction of the staircase that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The staircase shall be designed and constructed so that no part will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The staircase shall be constructed of lightweight, high strength-weight ratio material.

3.4.1 Adjustments and repairs. The staircase shall be constructed so that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases without the use of special tools.

* 3.4.2 Air transportability. The staircase shall be designed to be air transportable in type C-54 and larger cargo aircraft in accordance with MIL-A-8421.

3.4.3 Adjustable height. The staircase shall be so designed and constructed that the top platform is adjustable to all heights above ground level within the range of 85 to 145 inches. The adjustment provision shall be such that the top platform will mate with aircraft loading doors at any height within the required operating range without forming a height variance of more than ±3/8 inch between the doorsill of the aircraft and the top platform.

3.4.3.1 Positioning purposes. For positioning purposes with respect to aircraft having rear loading clamshell doors, the staircase shall be so designed that the overall height of the staircase can be reduced to 105 inches, or less, above ground level. This may be accomplished by lowering the staircase so that the platform height is only 69 inches above ground level, or by the use of handrail sections that can be easily and quickly folded or collapsed.

3.4.4 Drainage. The staircase shall include provisions for the continued drainage of water from the steps, platform, structural members and other components.

3.4.5 Tow bar storage. Adequate means shall be provided for stowing the tow bar in a suitable, out-of-the-way location on the staircase frame.

3.4.6 Mobility. The staircase shall have a high degree of mobility over rough concrete and smooth sod with emphasis on ease of manual positioning about the aircraft. Minimum mobility requirements for the staircase shall be those specified for type I, class 2 equipment of MIL-M-8090.

3.4.7 Lubrication. Suitable lubrication facilities as required to insure compliance of the staircase with the requirements specified herein shall be provided. Lubrication fitting shall be in accordance with MIL-F-3541.



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3.5 Performance. The staircase shall be capable of withstanding the following conditions:

a. A temperature of from -65° to +160° F.

* b. Dust as encountered in desert area.

c. Exposure to salt-sea atmosphere.

d. Wind velocities up to 60 Miles Per Hour (MPH) from any direction.

3.5 Roadability.

3.5.1.1 Turning. The staircase shall be capable of executing a minimum of 50 left-circle turns and 50 right-circle turns, both at maximum cramping angles, at a speed of 5 MPH (see 4.5.9).

3.5.1.1.1 Sudden stops. The staircase shall be capable of withstanding 25 sudden stops while undergoing the test specified to demonstrate 3.5.1.1.

3.5.1.2 Obstructions. The staircase shall be towable over 1-inch-high obstructions, regularly spaced at 6-foot intervals and at right angles to its direction of travel.

3.6 Lift mechanism. The lift mechanism shall be operated manually through mechanical or hydraulic means. It shall be operated from ground level by means of a conveniently located hand-operated crank or lever, with a maximum handle effort of 40 pounds. The lift mechanism shall be designed to attain an infinite number of positions throughout its operating range and shall be protected against atmospheric conditions and possible physical damage. The lift mechanisms shall conform to the test specified in 4.5.3.2.

3.6.1 Hydraulic system. When a hydraulic system is utilized, the following requirements shall apply.

3.6.1.1 The hydraulic system shall be designed to withstand a handle force of 200 pounds without leakage. A relief valve with a suitable setting shall be incorporated. The ram shall be fitted with O-type rubber packings in accordance with MIL-P-5315 and AN6227 and protected with backup rings in accordance with Drawing 49B6412 or 49B6413. A seal to prevent entrance of moisture into the cylinder shall be installed. All hydraulic fittings and lines (rubber or metal) shall be of an approved USAF high-pressure type suitable for operation in the temperature range of from -65° to +160° F. Nonflexing metal, hydraulic lines shall be used where flexing is not encountered in the operation of the staircase. Rubber or synthetic rubber hose shall be used where flexing is concerned.

3.6.1.1.1 Hydraulic tank. The hydraulic tank shall be equipped with a suitable filler neck on the top, a drain plug at the bottom, and an air vent in the filler cap. Baffles shall be provided in the tank to prevent the oil from splashing out when the staircase is moved. A suitable oil strainer shall be installed in the hydraulic suction line. The hydraulic system and tank shall be filled with oil conforming to MIL-H-5606.



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3.7 Lift lock. A device shall be provided to automatically lock the staircase in any ascent or descent position attained by the lifting mechanism. The lock shall withstand the test specified in 4.5.6.

* 3.8 Running gear. Four pneumatic-tired casters conforming to MIL-C-4751 shall be provided. Two MS24373-8WSC swivel casters shall be provided at the rear, or aircraft end, of the staircase. Swivel locks shall be provided. Two MS24373-8RSC rigid casters shall be provided at the front end of the staircase.

3.9 Immobilizing jacks. A minimum of four immobilizing jacks, one at each corner of the base, shall be provided to hold the staircase rigid during all loading and unloading operations. The jacks shall be designed to be in use at all platform elevations within the staircase operating range. The jacks shall compensate for uneven terrain and local depressions not exceeding 2 inches above or below the normal ground line. With a static load of 400 pounds placed on every other step, the immobilizing jacks shall be capable of supporting the staircase while it is raised and lowered through its operating range or positioned at any platform height within the range. The jacks shall be capable of supporting the fully elevated staircase with a static load of 500 pounds placed upon the top platform. The jacks shall withstand the test specified in 4.5.5.1.

3.9.1 Retracting mechanism. The jacks shall be provided with a retracting mechanism so designed that manual release of the ground jack pressure will cause the jack to fully retract without further attention of the operator. Manual release of the ground jack pressure shall require an operator effort of 40 pounds, or less.

3.9.2 Base plates. Base plates having a minimum area of 100 square inches shall be provided at the point of contact of the jack and the ground. Each plate shall swivel 10 degrees in all directions from the horizontal and shall clear the ground by a minimum of 4 inches when fully retracted.

3.9.3 Covering. The casters, hydraulic containers, and immobilizing jacks shall be covered in a suitable manner to present a pleasing appearance to the eye.

3.10 Handrails. Handrails shall be provided on each side of the staircase. They shall extend from the base of the stairs to the aircraft end of the top platform. The handrails shall maintain a constant height of 35 $\pm$1 inches above the steps and top platform at all platform heights within the operating range of the staircase. Handrail height above the steps shall be measured along a vertical line from the handrail to the nose of each step.

3.10.1 Accessibility. In order that aircraft doors may be opened and fleet servicing equipment used with the staircase in position, the handrails shall be so designed and constructed that they can be folded, collapsed, removed, or rotated to provide ready access to either side of the platform.

3.10.2 Siding. Solid, durable siding shall be provided to close the lower portion of the gap between the handrails and steps. The siding shall extend upward a minimum of half the distance between the stairs and handrails.

3.11 Electrical system.



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3.11.1 Batteries. The electrical system shall utilize two batteries conforming to type 6TN of MIL-B-11188, which shall be connected in series to provide a 24-volt power system. One filling of electrolyte shall be provided in a separate container for each battery.

3.11.1.1 Battery housing and mounting. The batteries shall be housed in an enclosed box constructed of suitable material to insure compliance with the requirements specified herein. The box shall be mounted in a convenient and protected location and shall be fitted with a quick-release cover to facilitate ease of inspection and servicing. The box and cover shall adequately protect the battery against atmospheric conditions and short circuiting of the electrical system.

3.11.2 Clearance lights. The staircase shall be equipped with amber-colored clearance lights located at the upper platform and ground end. They shall permit visibility from any position around the staircase. The lights shall be of the outdoor type and shall be weatherproof. Each light shall have a minimum illumination of 6 candlepower.

3.11.3 Step lights. The steps shall be illuminated to a sufficient degree that every step and the platform will be clearly visible at night to personnel ascending and descending the staircase. The lights shall be of the weather-proof type. The lights shall be recessed flush with the side of the staircase, and shall provide a minimum illumination of 6 candlepower over the entire step and platform areas.

3.11.4 Switch. One switch shall operate both sets of lights and shall be located in a convenient position near the lifting lever. The switch shall be of adequate rating for the power load imposed and shall be of outdoor weather-proof construction.

3.11.5 Wiring. The electrical wiring shall be standard commercial wire of adequate rating for the power load imposed and shall be of weatherproof construction.

3.12 Steps and top platform. Each step shall be 36 ±2 inches wide with a 12 ±1/2-inch tread and an 8 ±1/2-inch rise. The step treads shall not deviate more than ±1 1/2 degrees from the horizontal plane in any position. The step treads shall be coated with type II nonslip material conforming to MIL-W-5044, applied in accordance with MIL-W-5050. The step riser shall be covered and constructed in such a manner that a constant dimension will be maintained regardless of platform height. The top step shall be made in the form of a platform and a suitable bumper included thereon to prevent damage to the aircraft. The width of the platform shall substantially conform to the width of the steps and shall be 60 ±2 inches in length. The steps and platform shall withstand the tests specified in 4.5.3 and 4.5.4.

3.13 Towbar. A towbar shall be provided. It shall be designed to permit rapid attachment and detachment and shall swivel in the vertical plane. The towbar shall be provided with a lunette eye conforming to Drawing B7041347. Towbar attachment fittings shall be located at the rear, or aircraft end, of the staircase.



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* 3.14 Interchangeability. All parts having the same manufacturer's part numbers shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

3.15 Bolts, nuts, and pins. All bolts, nuts, and pins used in the construction of the staircase shall be cadmium plated.

* 3.16 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

* 3.17 Finishes and protective coatings. All exterior surfaces that are normally painted in commercial practice shall be treated and painted in accordance with MIL-T-704. The color shall be specified under 6.2.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 Workmanship.

3.19.1 General. The staircase, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. It shall be free from blemishes, defects, burrs, and sharp edges, and be accurate in every detail.

3.19.2 Riveting. Riveting operations shall be carefully performed to insure tightness and satisfactory heading of rivets.

3.19.3 Cleaning. The staircase shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of staircases shall be classified as follows:

a. Preproduction tests (see 4.3).

b. Acceptance tests (see 4.4).

4.3 Preproduction tests.

4.3.1 Sampling instructions. The contractor shall subject one staircase to the tests specified in 4.3.4.



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* 4.3.2 Test report. After the contractor completes the preproduction tests, he shall prepare a preproduction test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.3.3 Test sample for the procuring activity. Along with the preproduction test report, the contractor shall submit the staircase specified in 4.3.1, after testing, to the procuring activity for a review of the mechanical construction of the product and to perform any tests included in the specification after reviewing the contractor's test report.

4.3.4 Tests. Preproduction tests shall consist of all the tests specified in 4.5, conducted in the same order.

4.4 Acceptance tests. Acceptance tests shall consist of individual tests and sampling tests.

4.4.1 Individual tests. Each staircase shall be subjected to the following tests:

a. Examination of product (see 4.5.1).

b. Functional test (see 4.5.2).

c. Disassembly and reassembly inspection (see 4.5.10).

4.4.2 Periodic sampling tests. One staircase shall be selected at random from each 50 staircases or fraction thereof on the order and subjected to the following tests:

a. Static load (see 4.5.3).

b. Stability (see 4.5.4).

c. Jack test (see 4.5.5).

d. Lock test (see 4.5.6).

4.4.2.1 Rejection and retest. When one staircase selected from a production run fails to meet the specification, staircases still on hand later produced shall not be accepted until the extent and cause of failure are determined.

4.4.2.2 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of staircases on hand or produced later shall not be made until it is determined that staircases meet all the requirements of the specification.

4.4.2.3 Defects in staircases already accepted. The investigation of a test failure could indicate that defects may exist in staircases already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test methods.



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4.5.1 Examination of product. The staircase shall be inspected to determine compliance with the applicable drawings and to the requirements of this specification. Particular attention shall be given to the material requirements, dimensions, special processes, workmanship, finishes, and marking.

4.5.2 Functional test. All components, fixtures, controls, and mechanisms shall be operated, actuated, and moved. Malfunctioning, excessively rough operation, excessive vibration, or irregular movement shall be cause for rejection.

4.5.3 Static load. The staircase shall be completely assembled and a static load of 400 pounds placed at or near the center of every other step, including the top platform. Upon completion of the testing, evidence of failure, fracture or permanent deformation shall be cause for rejection.

4.5.3.1 With the staircase unloaded, it shall be operated through its entire range 100 times. Upon completions of the tests, the staircase shall be inspected. Failure of welds, materials, or bolts; elongation of boltholes, leakage of hydraulic fluid (if applicable), or major malfunction shall be cause for rejection.

4.5.3.2 The lift mechanism shall be inspected with the staircase in its fully extended position and a load of 500 pounds on the top platform. The staircase shall be allowed to remain under load for a period of 1 hour without the locking device in use. The settling rate of the top platform under this test load shall not exceed 0.125 inch in 2 hours. If the lift mechanism is hydraulically operated, tests shall be conducted to determine conformance to the applicable requirements of 3.6.1.1.

4.5.3.3 With the staircase assembled, a static load of 400 pounds shall be individually applied to each step and 500 pounds to the top platform. The load shall be placed at or near the center of the step and shall cover an area not to exceed 25 square inches. No point on the steps or top platform shall be depressed nor deflected more than 1/4 inch, and there shall be no failure nor permanent deformation.

4.5.4 Stability. With the immobilizing jacks extended, the maximum horizontal deflection at the platform level of the unloaded structure shall not exceed 2 inches when subjected to a horizontal load of 75 pounds applied normal to each side of the platform in turn. Under a distributed vertical load of 1,000 pounds, maximum vertical deflection of the platform shall not exceed 2 inches at any level from 85 inches to 145 inches.

4.5.5 Jack test. With the staircase in its fully extended position and static loads applied as specified in 4.5.3 and 4.5.3.2, the immobilizing jacks shall be fully extended to lift the entire weight of the staircase. Upon completion of the testing, there shall be no evidence of fracture, failure, permanent deformation, or binding.

4.5.5.1 With the test loads removed and the immobilizing jacks fully extended the staircase shall be subjected to a drawbar pull sufficient to move it a distance of 5 feet on smooth, dry concrete. The jacks shall show no evidence of permanent deformation nor failure.



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4.5.5.2 With the test loads removed and the immobilizing jacks fully extended, the retracting mechanism shall be manually actuated to release ground pressure. The jacks shall fully retract without further attention.

4.5.6 Lock test. With the staircase loaded as specified in 4.5.3, the staircase shall be operated through its range and stopped at various positions and the lock engaged. The lock shall hold the staircase in position without any tendency to slip.

4.5.7 Environmental tests.

* 4.5.7.1 High temperature. This test shall be conducted in accordance with the high temperature tests, Procedure I, of MIL-STD-810. The tests specified in 4.5.3 and 4.5.3.1 shall be performed while at 160° F.

* 4.5.7.2 Low temperature. This test shall be conducted in accordance with the low temperature tests, Procedure I, of MIL-STD-810. The tests specified in 4.5.3 and 4.5.10 shall be performed while at -65° F. For this test, utilize hydraulic fluid conforming to MIL-H-81019.

* 4.5.7.3 Dust. Hydraulic systems and other operating components of the staircase shall be tested in accordance with the sand and dust tests, Procedure I, of MIL-STD-810. At the completion of this exposure, the tests specified in 4.5.3 and 4.5.3.1 shall be repeated.

* 4.5.7.4 Salt fog test. Hydraulic systems and other operating components of the staircase shall be tested in accordance with the salt fog test of MIL-STD-810. At the completion of the exposure period, the tests specified in 4.5.2 and 4.5.10 shall be repeated.

4.5.8 Wind test. While in the fully extended position, the staircase shall be subjected to a simulated wind load having a velocity of 60 MPH. The loading shall occur normal to either side of the staircase. The staircase shall remain stable and upright.

4.5.9 Roadability. The staircase shall be tested as specified for type I, class 2 equipment in MIL-M-8090.

4.5.10 Disassembly and reassembly inspection. After completion of the foregoing tests, the staircases shall be completely disassembled and inspected for failure of hydraulic packings. The staircase shall then be reassembled, inspected, and checked for normal operation.

4.5.11 Base plate tests.

4.5.11.1 It shall be demonstrated that each base plate will swivel 10 degrees in all directions from the horizontal.

4.5.11.2 It shall be demonstrated that each base plate has a minimum ground clearance of 4 inches when fully retracted.

4.5.12 Packaging, packing, and marking. Preparation for delivery shall be examined for conformance with section 5.



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* 5. PACKAGING

5.1 Preservation and packaging.

* 5.1.1 Level A and B. Staircases shall be preserved in accordance with MIL-STD-162. Packaging materials which come in direct contact with unprotected surfaces susceptible to damage by corrosion shall conform to materials specifications referenced in MIL-STD-794 cited elsewhere in this document.

* 5.1.2 Level C. Staircases shall be preserved and packaged to protect material against physical damage and deterioration during favorable conditions of shipment, limited handling and short term storage. This may be the manufacturer's commercial practice if such meets the requirements of this level.

* 5.1.3 Commercial packaging. Staircases shall be preserved and packaged in accordance with MIL-STD-1188.

5.2 Packing.

* 5.2.1 Level A. Staircases preserved and packaged as specified in 5.1.1 shall be packed in overseas type exterior shipping containers. Selection of appropriate containers shall be limited to those listed in table I, MIL-STD-794. Containers shall be closed and strapped in accordance with the applicable container specifications or appendix thereto.

* 5.2.2 Level B. Same as level A except domestic type exterior shipping containers shall be used. Selection of appropriate containers shall be limited to those listed in Table I, MIL-STD-794.

5.2.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall be in accordance with Uniform Freight Classification Rules, or regulations of other common carriers, as applicable to the mode of transportation.

* 5.3 Commercial packing. Staircases preserved and packaged as specified in paragraph 5.1.3 above, shall be packed in accordance with the requirements of MIL-STD-1188.

* 5.4 Marking. Interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type MA-1 staircase covered by this specification is intended for loading and unloading passengers from military aircraft.

* 6.2 Ordering data.



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* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Levels of preservation and packaging, and packing (see section 5).

Level B preservation and packing is intended to provide economical but limited protection and should be specified only when it is determined to be appropriate. Effective period of protection is 6 months from date of initial packaging.

c. Where the preproduction test samples should be sent and instruction concerning the submittal of the test reports (see 4.3).

* d. Color (see 3.17).

6.2.2 Drawings. Invitations for bids should require the bidder to furnish drawings covering the general stairs, stair chassis, mechanical components, control, and other such items or features as will be required to clearly portray the complete unit offered and to substantiate conformance to the specification requirements.

* 6.2.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

(Project 1730-F258)

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-26851B | Staircase, Aircraft Boarding, Mobile, TYPE MA-1 |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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